CAN Signal Matrix — Display CAN, Pioneer Standard CAN and V2

Status: Working evidence matrix

Last updated: 2026-09-14

Audience: Firmware developer, decoder reviewer and hardware tester

Purpose and evidence levels

This document separates signals by physical bus and vehicle/profile source. It must not be read as proof that similarly named values are interchangeable.

Status	Meaning
Confirmed by MOT	Reproduced during a controlled live capture on the Microlino Pioneer
Plausible	Raw values and behaviour fit the proposed interpretation, but meaning or scale still needs an independent check
Externally supplied as V2	Supplied by a separate reverse-engineering source under the V2 label, but the exact vehicle, battery and BMS generation are not documented
Externally indicated	Present in supplied notes/screenshots, but byte order, scale, signedness or exact semantics remain incomplete
Not observed	Targeted ID did not appear during the stationary Pioneer capture
Observed on large-battery vehicle	Seen by MOT on a newer Microlino with the 200-240 km battery; interpretation still requires controlled state changes
Confirmed for large-battery decoder	Original PID/Flespi material and controlled live tests agree with physically plausible values and repeatable vehicle-state transitions

All MOT captures used two ESP32-C6 TWAI controllers at 500 kbit/s in listen-only mode. CAN1 was connected to Standard CAN and CAN2 to Display CAN.

Cross-bus signal summary

Signal	Display CAN	Pioneer Standard CAN (V1 vehicle)	Externally supplied / newer large-battery vehicle	Current assessment
Display SOC	`0x602 data[0] / 2`	No equivalent confirmed	External source: `0x18D data[7]`; large-battery live frame: `0x05` with meaning unknown	Display CAN confirmed by existing decoder. The external V2 assignment cannot be generalized to the tested large-battery vehicle.
Standard/BMS SOC candidate	Not used	`0x48D data[6]` internal candidate; `data[7]` display-correlated SOC	`0x1B1 data[0..1] / 100`, big-endian	Pioneer fields confirmed as separate whole-percent values; internal semantics remain provisional. Large-battery scan contains no `0x48D`.
Speed	`0x602 data[1] / 2`	Not found	No supplied mapping	Display-CAN decoder field.
Odometer	`0x602 data[4..7]` with current packed scaling	Not found	No supplied mapping	Display-CAN decoder field.
HV pack voltage	Not currently decoded	`0x18D data[3..4]`, unsigned little-endian mV	External source: same mapping; large-battery frame decodes to impossible 792 mV	Confirmed only on Pioneer. The supplied mapping is unusable on the tested large-battery vehicle.
Battery current	Not currently decoded	`0x18D data[1..2]`, signed little-endian; `raw x 0.3 A`	`0x2BA data[0..1]`, signed big-endian; raw / 10 A	V1 scale/sign confirmed for charge, traction and regeneration. V2 scale and charge direction confirmed at 6 A and 8 A; traction/regeneration validation remains open.
DC charging power	Dashboard power byte is unscaled and bidirectional	Derived as pack voltage × battery current	Derived from `0x1B0` voltage and `0x2BA` current	V2 produced approximately 1.18-1.30 kW DC at 1.34 kW AC and 1.36-1.53 kW DC at 1.55 kW AC; conversion loss and vehicle load explain the difference.
Plugged	Existing candidate `0x604 data[4] & 0x10`	`0x18D data[6]`: stable `0x10` unplugged and `0x20` plugged	`0x60D data[1] & 0x08`; corroborated by `0x610 data[0]`: `0x01` unplugged, `0x05` plugged-idle, `0x02` charging	V1 and V2 Standard-CAN states confirmed by repeated powered and unpowered plug transitions. Display-CAN is ignored when either Standard-CAN profile is active.
Active charging	Current `0x603` power-threshold inference is not reliable	Positive signed `0x18D data[1..2]`; idle offset about -3/-4	`0x610 data[0] == 0x02`; `0x05` is plugged-idle and `0x01` unplugged	V2 confirmed across unplugged, plugged-idle, active 6 A/8 A charging, charge stop and unplug.
Cell values A/B	Not decoded	`0x4AD data[0..1]` and `[2..3]`, unsigned little-endian mV	External source: same mapping; `0x4AD` absent on tested large-battery vehicle	Plausible on Pioneer. External vehicle assignment is unresolved and the mapping is unusable on the tested vehicle.
Cell min/max/delta	Not decoded	Derived from the two `0x4AD` values	External source: same derivation; unavailable without `0x4AD` on large-battery vehicle	Plausible only for the observed Pioneer payload.

Signal	Display CAN	Pioneer Standard CAN (V1 vehicle)	Externally supplied / newer large-battery vehicle	Current assessment
SOH	Not decoded	Not found	`0x1B1 data[2..3] / 100` candidate	83 identical large-battery frames decode to plausible 97.01%; external validation still pending.
Pedal/power request	`0x603 data[4]` is the dashboard power display	`0x37F data[2..3]` candidate	Same candidate	`0x37F` was present but constant in stationary Pioneer testing; driving evidence is required.

Pioneer Standard-CAN findings

CAN ID `0x18D`

Bytes	Type and conversion	Proposed meaning	Pioneer evidence	Relation to externally supplied V2 data
`data[0]`	`uint8`	rolling counter or sequence	Covered nearly the full byte range during stable charging; not a direct measurement	Unknown
`data[1..2]`	`int16`, little-endian	battery current; amperes = raw x 0.3	Scale and sign confirmed during charging, traction and regeneration	Likely related, but the V2 source does not expose an unambiguous raw-ID association
`data[3..4]`	`uint16`, little-endian mV	HV pack voltage	Confirmed by MOT	Same mapping was supplied, but it fails on the tested large-battery vehicle
`data[5]`	`uint8`	unknown	Stable `0x00` in recorded phases	Unknown
`data[6]`	`uint8` state	`0x10` unplugged; `0x20` plugged	Confirmed by repeated powered and AC-deenergized plug tests; transient `0x04` also occurred	V2 source calls it a charge/status byte
`data[7]`	`uint8`	Standard/BMS SOC candidate	Raw value confirmed; stayed 90 while Pioneer display showed 97–100%	Same byte was supplied as Display SOC; large-battery live value `0x05` remains uninterpreted

CAN ID `0x48D` (Pioneer only)

Bytes	Decoding	MOT topic	Evidence
`data[6]`	whole percent	`bms/soc_internal`	Separate internal SOC-like value; followed charge/discharge but remained about 6-8 points below display SOC
`data[7]`	whole percent	`bms/soc_display`	Matched visible Pioneer SOC exactly through controlled driving, rest and charging

Charging evidence for `0x18D data[1..2]`

Vehicle/charger state	Signed raw	Pack voltage	External measurement	Interpretation
Awake, unplugged	about -4	about 53.0 V	0 W	Idle offset
Plugged, EVSE AC-side off	about -4	about 53.0 V	0 W	Plugged but not charging
Stable charge	112–118	about 53.3 V	1,836 W gross AC input	`115 × 0.3 A × 53.3 V ≈ 1,839 W`
Taper charge at 98–99% display SOC	43–47	about 53.2 V	762 W gross AC input	`45 × 0.3 A × 53.2 V ≈ 718 W` before conversion losses
External AC removed, cable retained	about -4	falling from charge voltage	0 W	Charging stopped, still plugged
Natural completion at 100% display SOC	fell from +45 to -3	about 53.24 V at end	Fell from 762 W to 0 W	Stable post-charge idle offset

Lower-SOC measurements at several charge settings confirmed 0.3 A per raw unit. At about 50 V, the derived DC values tracked independent 1,809 W, 2,088 W and approximately 2,340 W references within expected conversion and vehicle-load differences. Three controlled road captures then confirmed negative current during traction and positive current during regeneration. Observed results included about 4.8 kW pedal-release regeneration, 7.8 kW with light braking and a later normal-road peak of about 20.9 kW discharge.

CAN ID `0x4AD`

Bytes	Type and conversion	Proposed meaning	Status
`data[0..1]`	`uint16`, little-endian mV	cell value A	Plausible on Pioneer; externally supplied under an unresolved V2 label
`data[2..3]`	`uint16`, little-endian mV	cell value B	Plausible on Pioneer; externally supplied under an unresolved V2 label

Bytes	Type and conversion	Proposed meaning	Status
derived	`min(A, B)`	minimum cell value	Plausible
derived	`max(A, B)`	maximum cell value	Plausible
derived	`max(A, B) - min(A, B)`	cell delta	Plausible

Stable examples included 4,081/4,078 mV before charging and approximately 4,105/4,100 mV during charge. Larger transient divergence occurred around charge start, so the exact semantics must remain provisional until repeated captures exclude multiplexing or transition-specific payload use.

Newer Microlino with large battery

A live Standard-CAN capture from a newer Microlino with the approximately 200-240 km battery received hundreds of thousands of CAN1 frames without bus errors. `0x18D` was present continuously, while `0x4AD` was absent. One stable `0x18D` payload was:

```
01 02 11 18 03 00 03 05
```

Applying the Pioneer layout would produce 1,306.2 A, 792 mV pack voltage, status `0x03` and an SOC candidate of 5%. The current and voltage results are physically impossible, proving that the Pioneer byte layout cannot be used for this vehicle. The frame remains raw discovery data; controlled tests found the V2 plug and charging states on separate IDs `0x60D` and `0x610`. Traction and regeneration validation of the V2 current signal remains open.

The earlier source may have been recorded from a Pioneer/V1 vehicle, or from a newer Microlino whose medium battery and BMS retain the Pioneer layout. This is a working hypothesis only: the source does not document the exact vehicle, battery or BMS generation. The name V2 must therefore not be treated as a single CAN protocol generation.

Confirmed large-battery BMS layout

The recovered original PID workbook and Flespi configuration identify a separate big-endian layout. A 16,675-frame capture from the large-battery vehicle contains all three IDs with matching physically plausible values:

CAN ID	Direct bus decoding	Captured evidence	Decoder status
`0x1B0`	bytes 2-3 pack mV, 4-5 minimum cell mV, 6-7 maximum cell mV; big-endian	47,317-47,326 mV pack and 3,635-3,643 mV cells	Implemented
`0x1B1`	bytes 0-1 SOC x100, bytes 2-3 SOH x100; big-endian	32.15% SOC and 97.01% SOH	Implemented as BMS-specific values
`0x2BA`	bytes 0-1 signed big-endian current x10	stationary -1.9 to -2.1 A and +21.9 to +28.8 A while charging	Implemented; charging sign confirmed, traction/regeneration pending
`0x60D`	byte 1 bit 3 (`0x08`) set while cable connected	repeated unplugged/plugged transitions with EVSE energized and de-energized	Implemented and confirmed
`0x610`	byte 0 state: `0x01` unplugged, `0x05` plugged-idle, `0x02` actively charging	repeated plug, 6 A/8 A charge, stop and unplug transitions	Implemented and confirmed

The direct CAN order must be used by MOT. The Flespi `manual.can.hex.*` conversion reverses groups of 16-bit words internally and must not be copied as bus byte order. Pack power is derived only when fresh `0x1B0` voltage and plausible `0x2BA` current are available. Plugged and active charging come from the confirmed `0x60D`/`0x610` states above. AWS continues to publish Display-CAN `0x602` as `display/soc`; BMS SOC is kept separate and never overwrites the confirmed display value.

Other targeted Pioneer IDs

CAN ID	Pioneer result	External V2 information
`0x101`	Not observed	Byte 4 candidate for positive power request; unit/scale incomplete
`0x107`	Not observed	Separate ESPHome source watched bytes 0–3 without a decoding rule
`0x1B0`	Not observed	Confirmed big-endian pack/min/max layout on the large-battery vehicle
`0x1B1`	Not observed	Confirmed big-endian SOC/SOH x100 layout on the large-battery vehicle
`0x2BA`	Not observed	Signed big-endian current /10; charging sign confirmed, traction/regeneration pending
`0x60D`	Not observed	Byte 1 bit 3 is the confirmed physical plug-presence signal
`0x610`	Not observed	Byte 0 is confirmed as `0x01` unplugged, `0x05` plugged-idle, `0x02` charging
`0x37F`	Observed and constant while stationary	Bytes 2–3 pedal/power-request candidate

Display-CAN signal map

CAN ID `0x602`

Bytes	Current decoding	Status
`data[0]`	SOC = byte / 2	Implemented Display-CAN signal
`data[1]`	speed = byte / 2 km/h	Implemented Display-CAN signal
`data[4..7]`	packed odometer / 1024 km	Implemented Display-CAN signal
derived	estimated range from SOC	Firmware-derived estimate, not a direct vehicle signal

CAN ID `0x603`

The external source uses one-based byte and bit numbering. Positions below are converted to zero-based firmware indices, assuming its bit 1 is the least- significant bit.

Firmware position	Mask	Meaning	Status
`data[1]`	`0x01`	seat-belt sensor/status	Externally indicated
`data[2]`	`0x08`	handbrake	Externally indicated
`data[2]`	`0x02`	parking/position light	Externally indicated
`data[2]`	`0x04`	low beam	Externally indicated
`data[3]`	`0x80`	D/R interlock warning when brake was not pressed first	Externally indicated
`data[3]`	`0x20`, `0x40`	indicator-state bits	Externally indicated; left/right mapping missing
`data[4]`	full byte	dashboard power display	Implemented; sign/scale and charging interpretation remain provisional
`data[5]`	`0x01`	sport mode	Externally indicated
`data[6]`	`0x02`	brake pressed	Externally indicated

`data[4]` is a bidirectional dashboard display value. It must not by itself be used as proof of active charging. The current firmware threshold-derived charging state is compatibility behaviour pending replacement by the confirmed Standard- CAN current/plug signals or another independently verified source.

CAN ID `0x604`

Firmware position	Mask	Meaning	Status
`data[4]`	`0x10`	plugged candidate	Implemented, but not yet directly compared with the controlled plug sequence

Decoder policy

- Preserve Display-CAN and Standard-CAN values as source-specific telemetry.
- Do not overwrite Display-CAN SOC with `0x18D data[7]` on Pioneer.
- Permit every registered decoder profile on every physical CAN controller; board defaults are configuration policy, not decoder-engine restrictions.
- Keep CAN operation listen-only during discovery and pilot operation.
- Promote plausible or externally indicated fields to confirmed only after a repeatable state change or independent physical reference.

Remaining validation

- Stationary/drive comparison of `0x37F` to test the pedal hypothesis.
- Direct Display-CAN `0x604` plug-state comparison.
- Compare `0x2BA` during traction and regeneration; its charging direction and the separate large-battery plug/charge states are now confirmed.

- Identify the exact vehicle, battery and BMS generation behind the externally supplied `0x18D`/`0x4AD` material.
- Controlled cross-channel injection and extended dual-CAN soak.

Related documents

- CAN and decoder pipeline
- Firmware architecture
- C6-001 dual-CAN sprint